PDE Ledger

Archive Ledger Skeleton

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Purpose and Scope

TODO. B3 will fill the rebuilt ledger purpose, source hierarchy, and scope boundaries.

0.1 The Role of This Ledger

Placeholder for the rebuilt ledger's role and audience.

0.2 Primary Source Hierarchy

Placeholder for the source hierarchy used by the rebuilt ledger.

0.3 Document Contract

Placeholder for the provenance, status, verification, and downstream-use contract.

0.4 Scope Boundaries

Placeholder for the limits of the rebuilt ledger.

0.5 Completeness Criterion

Placeholder for the completion standard used by later build phases.

How to Read This Ledger

TODO. B3 will fill the reader guide after the rebuilt stage and Part structure is set.

0.6 Layer Structure

Placeholder for the relationship between theorem-block prose, stage cards, notes, and audits.

0.7 Reading Paths

Placeholder for operational, verification, provenance, and authoring reading paths.

0.8 Citation Discipline

Placeholder for result anchors, theorem labels, equation labels, and stage provenance.

0.9 Stage Entries

Placeholder for the expected fields in future stage cards.

0.10 Gap Handling

Placeholder for demotion, splitting, open-gate, and failed-route handling.

Claim-Status Firewall

TODO. B3 will fill the rebuilt ledger's status rules and examples.

0.11 Allowed Status Tags

Status tag	Meaning
Exact	Placeholder.
Exact Within Closure	Placeholder.
Reduced / Controlled Reduction	Placeholder.
Effective Closure	Placeholder.
Numerically Located	Placeholder.
Open	Placeholder.

0.12 Audit-Status Vocabulary

Placeholder for audit evidence categories.

0.13 Status Inheritance

Placeholder for weakest-essential-input and status propagation rules.

0.14 Citation Language

Placeholder for safe status-aware citation wording.

Notation Firewall

TODO. B3 will fill notation rules after the rebuilt sectors and stage packets are fixed.

0.15 Global Anti-Ambiguity Rules

Placeholder for global notation safeguards.

0.16 Coordinate, Index, and Operator Conventions

Placeholder for coordinate, index, and operator conventions.

0.17 Field and Sector Variables

Placeholder for sector-specific variable conventions.

0.18 High-Risk Symbol Collision Table

Placeholder for symbol collision accounting.

0.19 Notation Checklist

Placeholder for future stage write-up checks.

Result Anchor Map

TODO. B3 will assign rebuilt result anchors after the Part and stage plan is settled.

0.20 Anchor Use Rules

Placeholder for stable citation-anchor rules.

0.21 Anchor Naming Convention

Placeholder for rebuilt anchor naming.

0.22 Top-Level Result Anchors

Anchor	Stage range	Citation target
No anchors assigned yet.		

0.23 Maintenance Checklist

Placeholder for anchor maintenance rules.

Dependency Map

TODO. B3 will fill the rebuilt dependency graph after stages are authored.

0.24 Arrow Types

Placeholder for dependency-arrow types.

0.25 Global Dependency Spine

Placeholder for the rebuilt ledger's dependency spine.

0.26 Top-Level Dependency Table

Block	Inputs	Outputs
No dependency blocks assigned yet.		

0.27 Revision Rules

Placeholder for dependency-map revision rules.

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Stage cards for Part II are collected in Appendix B.

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Stage cards for Part III are collected in Appendix C.

Appendix A

Part I Stage Cards

A.1 Stage 004: GNLS Action and Dimensional Foundation

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Fix the field content and $\{L, T, M\}$ dimensional dictionary of the GNLS parent action, derive the pin null-relation and healing scale, and land the base-system verdict `RETAIN_L_T_M`.

Status. Exact Within Closure. The dimensional dictionary, sound-speed and healing anchors, pin null-relation, and base-system verdict are exact symbolic results; the honest landing carries named residuals (mass non-emergence and the $\{L, T\}$-conversion gaps), deferred to pathA_21/22 and not repaired here.

Parent action and field content. The 4D-bulk medium is a GNLS condensate with order parameter ψ, number density $\rho = |\psi|^2$, and EOS $P = K\rho^5$ (from $U(\rho) = \frac{K}{4}\rho^5$):

$$\mathcal{L} \sim i\hbar\,\psi^*\partial_t\psi - \frac{\hbar^2}{2m_{\mathrm{GNLS}}}\,|\nabla\psi|^2 - U(\rho).$$

Dimensional dictionary (two tiers). **Primitive inputs.** $[\hbar] = ML^2T^{-1}$, $[m_{\mathrm{GNLS}}] = M$, $[\rho_0] = L^{-4}$ (state datum) — posted and consistency-checked against action usage. **Derived by composition.** $[\psi] = L^{-2}$, $[\rho] = L^{-4}$, $[K] = ML^{18}T^{-2}$, forced by action homogeneity across all 17 harness checks (14 foundation, including the bulk continuity equation, plus 3 $\{L, T\}$-representation gates).

Derived anchors. **Sound speed and healing length.**

$$c_{s0}^2 = \frac{5K\rho_0^4}{m_{\mathrm{GNLS}}}, \quad [c_{s0}] = LT^{-1}; \qquad h_0 = \frac{5K}{4}\rho_0^4 = \frac{m_{\mathrm{GNLS}}c_{s0}^2}{4}, \quad \xi_h = \frac{\sqrt{2}\,\hbar}{m_{\mathrm{GNLS}}c_{s0}}.$$

Pin null-relation. Four pins $\{a, c_{s0}, \hbar, m_{\mathrm{GNLS}}\}$ on three base dimensions give rank 3, nullity 1:

$$a = \frac{\hbar}{m_{\mathrm{GNLS}}c_{s0}}, \qquad a/\xi_h = 1/\sqrt{2}.$$

Verdict. **RETAIN_L_T_M.** Computed from two conjuncts: the $\{L, T\}$-rejection gate (dropping M leaves ≥ 1 non-dimensionless target, needing M-bearing conversion factors LT^2M, L^2T^2M, L^2T^2) *and* the mass fork (m_{GNLS} explicit of dimension M; m_{defect} not derived, `INFLOW_MASS_SOURCE_MISSING`). The $\{L, T\}$ projection is a natural-unit representation only, not a base-system change.

Carried residuals (first-class, not repaired). `INFLOW_MASS_SOURCE_MISSING` (m_{defect} emergence deferred to pathA_21; $\hbar J/c_\gamma^2 = M$ is dimensional-only), `LT_R_norm_gate_fails_without_new_conversion_factor` (`REJECTS_TRUE_LT_BASE`) and the two 4D/3D R-norm conversion gaps (pathA_22), plus `A_PIN_IS_BRANCH_MOMENT_N` `EOS_FROM_GNLS_FACTOR`, `NO_NET_ACCRETION_BC_UNDERIVED`, `M_TO_G_UNIFICATION`, `SCALE_MAP_INPUTS`.

A.2 Stage 005: EOS Sound Speed and Light/Sound Ratio as a Free Knob

Part / anchor. Part I – The medium

Claim status. Reduced / Controlled Reduction

Purpose. Derive the phonon sound speed from the imposed EOS, pin the three velocity scales and the $c = c_\gamma$ wave-sector ceiling, and land the light/sound ratio $\lambda_\gamma = c_\gamma/c_s$ as a free calibration input (`C_GAMMA_RATIO_UNDERDETERMINED`).

Status. Reduced / Controlled Reduction. The sound-speed law and the $c = c_\gamma$ ceiling are exact symbolic results *within the imposed EOS closure*; the honest headline is the CALIBRATED landing: $\lambda_\gamma = c_\gamma/c_s$ is *unpinned by the parent action* – a free calibration input, with the tail $(c/c_s)^3 = \lambda_\gamma^3$ carried symbolic (not set to 1). Consuming stage: it cites `ledger_stage004` (I-1)'s dictionary and `EOS_FROM_GNLS_FACTOR` handoff (h_0, ξ_h) and owns the sound-speed derivation I-1 deferred here.

Sound speed from the EOS. From the imposed $P = K\rho^5$ (`EOS_CLOSURE_IMPOSED`) and the barotropic definition, by symbolic differentiation:

$$c_s^2(\rho) = \frac{1}{m_{\text{GNLS}}}\frac{dP}{d\rho} = \frac{5K\rho^4}{m_{\text{GNLS}}}, \quad [c_s] = LT^{-1}, \qquad \frac{d\ln c_s}{d\ln\rho} = 2.$$

The factor 5 falls out of the derivative. EARNED relative to the imposed EOS.

Three speeds and the $c = c_\gamma$ ceiling. **Velocity scales.** $v_b = (\hbar/m_{\text{GNLS}})\nabla\theta$, c_s, c_γ, each of dimension LT^{-1} (v_b's pin consumed from I-1). **Ceiling (derived from the wave sector).** For the trapped mode $\omega = c_\gamma\sqrt{k^2 + k_\perp^2}$,

$$\frac{d\omega}{dk} = \frac{c_\gamma k}{\sqrt{k^2 + k_\perp^2}}, \qquad c_\gamma^2 - \left(\tfrac{d\omega}{dk}\right)^2 = \frac{c_\gamma^2 k_\perp^2}{k^2 + k_\perp^2} \geq 0,$$

strictly positive for $k_\perp \neq 0$: the group speed is bounded by c_γ and approaches it. The bound-mode clock $\omega_0 = c_\gamma k_\perp$ gives the boosted internal rate ω_0/γ (time dilation) with no $E = mc^2$ premise.

Coupled principal symbol (pathA_20b support). On the neutralized homogeneous background the coupled symbol block-diagonalizes: the Madelung phonon determinant $\det M = \hbar(\omega^2 - c_{s0}^2 k^2)$ with $c_{s0}^2 = 5K\rho_0^4/m_{\mathrm{GNLS}}$ falling out via $h' = 5K\rho_0^3$, the gauge block $C_E\omega^2 - C_B k^2 = C_E(\omega^2 - c_{\mathrm{bulk}}^2 k^2)$ ($c_{\mathrm{bulk}}^2 = C_B/C_E$), and $\det P = \det M \cdot P_T^2$. The off-diagonal current/London terms are lower order and do not set the cone (`BULK_PRINCIPAL_TRANSVERSE_BRANCH_ESTABLISHED`); $[c_s] = [c_\gamma]$ is labeled *non-evidentiary* for equality.

The free-knob landing and its negative control. C_GAMMA_RATIO_UNDERDETERMINED. Computed, not asserted. With $C_E, C_B, K, \rho_0, m_{\mathrm{GNLS}}$ independent, the residual $C_B/C_E - 5K\rho_0^4/m_{\mathrm{GNLS}}$ is not identically zero and no source equation is present, so `forced_equals_valid = False` $\Rightarrow$ `FORCED_EQUALITY_REJECTED_WITHOUT_SOURCE_EQUATION`. It is *reversible*: inserting the identity $C_B \to 5K\rho_0^4 C_E/m_{\mathrm{GNLS}}$ flips it to `C_GAMMA_EQUALS_C_S` – so `EQUALS` is reachable only via an actually-inserted source equation the action does not supply. Two-layer sharpening: bulk `C_GAMMA_BULK_UNDERDETERMINED`, brane `C_GAMMA_RATIO_STILL_UNDERDETERMINED`; if c_{bulk} is ρ_0-independent then $\lambda_\gamma \propto \rho_0^{-2}$ ($d\ln\lambda_\gamma/d\ln\rho_0 = -2$).

Carried residuals (first-class, not repaired). `EOS_CLOSURE_IMPOSED`; `C_GAMMA_RATIO_UNDERDETERMINED`; `STATIONARY_PROFILE_UNDERDETERMINED_BY_BRANCH_DATA`; `NO_NET_ACCRETION_BC_UNDERIVED`; `HBAR_PROVENANCE_U` `HBAR_FREE_SUBSTRATE_RELATION_MISSING`; `H_2PI_RATE_CLASSIFICATION_UNDERDETERMINED`; `BULK_METRIC_SPEED_` `PARENT_METRIC_ACOUSTIC_IDENTIFICATION_MISSING`; `BRANE_ZERO_MODE_REDUCTION_UNDERIVED`; `BRANE_PHOTON_CONE_REQUIRES_PROFILE`. The mass bridge $\hbar J/c_\gamma^2 = M$ is a labeled dimensional candidate only, not a m_{defect} derivation.

A.3 Stage 006: Two-Phase Material-State Ontology (Order Field χ_B)

Part / anchor. Part I – The medium

Claim status. Exact Within Closure

Purpose. Formalize the brane/bulk ontology as a POSTULATED two-phase action – one conserved medium, order field $\chi_B \in [0,1]$; brane = ordered shear-supporting phase, bulk = the same medium destructured, throat = phase-conversion site – dimensionally classified (`ACTION_SPECIFIED_CLASSIFIED`), with the recovery reduction to the frozen old-ledger S_{leak} verified assert-zero and the θ-as-Maxwell-φ no-go carried able-to-fail.

Status. Exact Within Closure. **The wall is made explicit as a postulated field – it is NOT derived from the one medium.** Every term of the free-energy functional and order-state balance is POSTULATED and labeled (13 pinned modeling choices P1–P13); the EARNED content is exact *within that imposed closure*: structural-closure identities, single-kink wall admission, and the recovery reduction against the frozen firewall. Cross-engine agreement proves consistent encoding of the same postulated F, not derivation. Consuming stage: cites `ledger_stage004` (dictionary, single-well U), `ledger_stage005` (c_{s0}^2), and `ledger_stage003` ($c_\gamma^2 = \mu_R/\rho_{br}$, $C_J = -J\rho_{B0}$, $B_{\mathrm{eff}} = \rho_{B0}^2/\chi_c$, the second-class$\Rightarrow$stray-DOF classification rule) with exact-value citation-integrity checks.

The postulated action and balance.

$$F = \int d^4X \left[\tfrac{1}{2} m_{\mathrm{GNLS}} n|\mathbf{u}|^2 + U(n) + f_B(\chi_B) + \tfrac{\kappa_B}{2}|\nabla_4\chi_B|^2 + \chi_B f_{\mathrm{shear}} + f_{\mathrm{throat}} + f_{\mathrm{mix}}\right],$$

$$\partial_t n + \nabla_4\cdot(n\mathbf{u}) = 0, \qquad \partial_t(\chi_B n) + \nabla_4\cdot(\chi_B n\mathbf{u} + \mathbf{J}_\chi) = n\Gamma_B, \quad [\Gamma_B] = T^{-1}.$$

Key pins: kinetic term carries the constituent mass (`KINETIC_MASS_FACTOR_PINNED`; $[n] = L^{-4}$); double-well $f_B = a_B\chi_B^2(1-\chi_B)^2$ with minima $\{0,1\}$, n-independent (`SAME_DENSITY_DEGENERACY_POSTULATED`) – required because the cited $U(\rho) = (K/4)\rho^5$ is *single*-well; MacCullagh-form shear gate on the displacement curl; $\chi_B \perp P$ (`POSTULATED_ANISOTROPY` dims-only); $\mathbf{J}_\chi = 0$ default; global returns $R_0 = -M_0$, $R_1 = -D_1$ printed as postulates, not asserted locally; throat = phase-conversion ontology, not a solve. Every integrand term audited to $ML^{-2}T^{-2}$; adjunct rows $(\mu_\chi, M_\chi, P_{\text{order}}, M_n)$ dimension-pinned, with the energy ledger fixed to the variationally consistent $P_{\text{order}} = \int \mu_\chi D_t\chi_B\, d^4X$ (`HANDOFF_P_ORDER_N_PLACEMENT_CORRECTED`).

Structural closure and wall admission (EARNED within closure). The advective form $D_t\chi_B = \Gamma_B - (1/n)\nabla_4\cdot\mathbf{J}_\chi$, the disorder complement for $n_D = (1-\chi_B)n$, and the Γ_B-cancelling sum reproducing total conservation are derived by real derivative expansion in both engines. The static EL equation $\kappa_B\chi_B'' = f_B'$ admits the kink $\chi_B = \frac{1}{2}(1+\tanh(w/2\delta))$ with solved width $\delta = \sqrt{\kappa_B/2a_B}$ and surface tension $\sigma_{\text{wall}} = \sqrt{2a_B\kappa_B}/6$, $[\sigma_{\text{wall}}] = ML^{-1}T^{-2}$. **Slab caveat (carried).** Bulk on both sides makes the brane a finite kink–antikink *slab*; the double-well selects no slab width – W_{slab} is an un-earned input mapping onto the old ledger's L/a self-selection item (sim-deferred). Wall admission $\neq$ slab stability.

Recovery reduction (EARNED within closure; the frozen firewall). Projection with unit-normalized $W(w)$ and in-engine IBP gives the two-source law $\partial_t\rho_B + \nabla_3\cdot\mathbf{J}_B = S_{\text{flux}} + S_{\text{convert}}$ with the general $\chi_B(w), \Gamma_B(w), J_\chi^w(w)$ computationally live; the in-engine limit $\chi_B \to 1, \Gamma_B \to 0, \mathbf{J}_\chi \to 0$ reduces it assert-zero to the *independently constructed* frozen target $S_{\text{leak}} = -[Wj^w]_\partial + \int W'j^w\, dw$, $j^w = nu^w$ (stage_243), including the stage_244 Gaussian one-mode closed form $S_{\text{leak}} = \sqrt{2}\,\mu_w\rho_0\mathcal{E}_0/(2\sqrt{\pi}\lambda^3)$ re-derived by direct integration (`RECOVERY_REDUCTION_VERIFIED`). Conditionality computed: $\chi_B = c \neq 1$, $\Gamma_B \neq 0$, $J_\chi^w \neq 0$ each leave nonzero computed residuals – the limit does real work.

The θ-as-Maxwell-φ no-go (carried FATAL_FLAW). A stable order-parameter phase has positive gradient energy ($\kappa_{\text{phase}} > 0$) $\Rightarrow$ $K_\theta = -\kappa_{\text{phase}} < 0$ in the Lagrangian; Maxwell's electric square needs $K_\theta = C_J^2/\rho_{br} > 0$ – opposite signs (exact predicate). The constraint bracket $\{\Phi_1, \Phi_2\} = k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})/\rho_{br}$ has its zero locus solved in-engine at exactly $K_\theta = +J^2\rho_{B0}^2/\rho_{br}$: via the consumed stage_003 rule the provenance branch lands `FAIL_CAUCHY_STRAY_LONGITUDINAL` (finite $B_{\text{eff}} = \rho_{B0}^2/\chi_c$), the $B_{\text{eff}} = 0$ branch `FAIL_C5_LONGITUDINAL_ZERO_MODE`, and the tuned fixture ($K_\theta = J^2\rho_{B0}^2/\rho_{br}, B_{\text{eff}} = 0, m_\theta^2 = 0$) `C5_RESOLVED_MAXWELL_BY_TUNING` – flagged `BY_TUNING`, never `WITH_PROVENANCE` (flag computed from provenance booleans; detunings re-fire the pathA_36 control tokens). The only provenance sign-flip ($\kappa_{\text{phase}} < 0$) is a Lifshitz instability, $k_*^2 = -\kappa_{\text{phase}}/2\kappa_4 > 0 \iff \kappa_{\text{phase}} < 0$ – the already-killed pathA_25 wall (`FAIL_LIGHT_STARVED` lineage). The transverse sector is untouched: the ε-parametrized dispersion derived in-engine gives $\omega^2 = (\mu_R/\varepsilon)k^2$; $\varepsilon = \rho_{br}$ reproduces the consumed c_γ^2 (`PASS_TRANSVERSE_UNDISTURBED`), $\varepsilon \neq \rho_{br}$ fires `FAIL_TRANSVERSE_DISTURBED`. **This stage does not earn light** – the no-go is the point.

Drift and carried items (first-class, not repaired). `DRIFT(6)` computed: $\{\chi_B$; a_B; κ_B; α_{aniso}; Γ_B; gating struct (rung-W reconciliation printed; `THETA_BRANCH_DEAD_NOT_ADMITTED`; ρ_{B0}, χ_c cross-referenced to the pathA_41 Part-VI drift trio, not double-counted). Carried: `SECOND_MEDIUM_DRIFT` lineage, the slab width W_{slab}, Gate-L $\delta w = u_w$ translation Goldstone (deferred), $\mathbf{J}_\chi \neq 0$, $f_{\text{throat}}/f_{\text{mix}}$ placeholders, dynamics/energy-ledger adjunct. Two source-prose dimensional defects pinned and ablation-documented (kinetic mass factor; P_{order} n-placement).

A.4 Stage 007: Shear-Surface G0 Freeze (Frozen Medium Action, Drift Ledger, DOF)

Part / anchor. Part I – The medium (closing Part-I stage)

Claim status. EXACT WITHIN CLOSURE

Purpose. Formal home of the pathA_35 G0 shear-surface *constitutive* freeze: the frozen brane/light action ($L_{\mathrm{Mac}}+L_{Pu}+L_{u_w}$ under the Gaussian profile $g_\ell(w)$), hash-guarded (`T0_SHEAR_FROZEN(d9520d3819c3)`), with the honest drift ledger `SECOND_MEDIUM_DRIFT_AT_FREEZE(11)` computed from an enumerated table and the flat-brane linear DOF$=8$ rank-computed.

Status. EXACT WITHIN CLOSURE. **The 11 freeze inputs are POSTULATED/CALIBRATED – the freeze freezes *terms*, not gate answers.** The EARNED content is exact within that declared closure: (i) *freeze fidelity* – both engines re-extract the canonical `freeze-action` blocks by byte-exact fence-parsing (never fixed offsets) and assert SHA-256 equality (G0 `d9520d3819c3f7...`, T0 `8fa41ac51e88...` byte-embedded); (ii) the *flat-brane linear DOF* $=8$ computed from rank/nullity bookkeeping (in-plane u^a curl-transverse 2 + kinetic–curl 1; T0 P^i tangent 3 + radial 1; u_w 1; C5 φ **0** – the absent longitudinal partner the later C5 crux attacks); (iii) the *dimensional firewall* over every frozen term. This stage computes **no** gate verdict (Gate-L excluded; exposure names carried as prose provenance only) and does **not** earn light (Stage 003 derives the two transverse photons at $c_\gamma^2=\mu_R/\rho_{br}$ from this stage's frozen L_{Mac}).

The frozen action. Kept (0 new): the GNLS parent action and the T0 polar-OP action (byte-for-byte, hash-embedded). Frozen brane blocks under $S_{\mathrm{brane}}=\int dt\, d^4X\, g_\ell(w)\,[L_{\mathrm{Mac}}+L_{Pu}+L_{u_w}]$:

$$L_{\mathrm{Mac}}=\tfrac{1}{2}\rho_{br}(\partial_t u^a)^2-\tfrac{1}{2}\mu_R\,\Omega_u^a\Omega_u^a,\qquad L_{Pu}=-\lambda_{Pu}\,\varpi_a\,\Omega_u^a,\quad \varpi_a:=(\hat w\times P_\parallel)_a,$$

$$L_{u_w}=\tfrac{1}{2}\rho_{br}(\partial_t u_w)^2-\tfrac{1}{2}\rho_{br}\Omega_w^2u_w^2,\qquad g_\ell(w)=\frac{e^{-(w/\ell_g)^2}}{\sqrt{\pi}\,\ell_g},\quad \int g_\ell\,dw=1\ \text{(derived)}.$$

Preserved verbatim: L_{Pu} *re-admits the* ϵ*-contracted/chiral class excluded by T0* and requires the conceded axis $\hat w$ – a structural-postulate cost, not a free choice.

The drift ledger (computed). Both engines enumerate the 11 and *compute* the subcounts and the verdict string (the source gate's hardcoded literals are retired): **4 constants** $\rho_{br}\,[ML^{-3}]$, $\mu_R\,[ML^{-1}T^{-2}]$, $\lambda_{Pu}\,[ML^{-1}T^{-2}]$, $\Omega_w\,[T^{-1}]$ (table dims asserted against the firewall targets); **1 function** $g_\ell(w;\ell_g)$ (fixed Gaussian, one width, target-blind admission); **6 structural postulates** (conceded $\hat w$+wall; free-slip u^a; P^i reused as Cosserat reservoir; massless spin-wave baseline; the parity-even P–u operator; no C5 φ). New-field content (u^a 3 + u_w 1 = 4 DOF) is counted *separately.* **Erratum (2026-07-04), carried first-class: the 11 STANDS** – no ρ_{br} overcount (pathA_25's $\varrho_{br}[\rho]$ belongs to the closed density-smectic candidate, `OUT_OF_ACTIVE_NG5`); $\{\rho_{br},\mu_R\}$ carry the registered-pending Route-A reduction (`ROUTE_A_UNDERDETERMINED_MISSING_NONLINEAR_THROAT`, register R10); the honest cross-sector drift $\{\rho_{B0},\chi_c,C_{hu}\}$ is a Part-VI (pathA_41) item, guarded in-engine against absorption into the 11.

Supersession and firewalls. Both facts asserted: the Stage 006 χ_B wall superseded the fixed-shape $g_\ell(w)$ as the *material-state* closure, while G0 *remains* the light-sector *constitutive* freeze (Stage 003 consumes L_{Mac} as-is) – a scope split (register R21), not a reduction. Notational firewall (register R22): μ_R (3D brane, $ML^{-1}T^{-2}$) and Stage 006's $\mu_R^{(4)}$ (4D density, $ML^{-2}T^{-2}$) are asserted dimensionally distinct, related only by the *pending* R17 projection $\mu_R = \int \chi_B\, \mu_R^{(4)} dw$ (dim-consistency asserted).

Verification. Dual-engine, independent routes (the source pair's `-compare` JSON payload mirror is retired): SymPy 96 PASS / Mathematica 94 PASS, both exit 0, CWD-independent. Able-to-fail teeth all fire: the five source-gate ablations (2 dimensional, 3 DOF each recomputing $7 \neq 8$), two hash teeth (in-block byte corruption; missing fence), five enumeration teeth (drop entry; miscategorize; inject ρ_{B0}; corrupt n pre-verdict; corrupt table dim), and the μ_R firewall tooth. Full tri-review on fresh agents: arbiter re-run, `FIDELITY_CLEAN` (block-by-block coverage diff, no dropped check), `ADVERSARIAL_CLEAN` (26-run mutation matrix: hash genuinely fence-parsed – one byte inside the block fails, one byte outside passes; dual-corruption of both engines fails both; no cross-engine comparison anywhere).

Appendix B

Part II Stage Cards

B.1 Stage 001: Solid-Angle and Second-Moment Primitives

Part / anchor. Part II – Gravity

Claim status. EXACT

Purpose. Derive the unit-sphere solid angles and normalized second moments used by the pathA_21c matter-stress force assembly.

Status. EXACT. This stage is closed geometry: no profile, normalization, or field-equation residual remains inside the stated claims.

Derivation. For $S^2 \subset \mathbb{R}^3$, take

$$n(\theta,\phi) = (\cos\theta, \sin\theta\cos\phi, \sin\theta\sin\phi), \qquad dS = \sin\theta\, d\phi\, d\theta.$$

Then **Omega2 solid angle.**

$$\Omega_2 = \int_0^{\pi}\int_0^{2\pi} \sin\theta\, d\phi\, d\theta = 4\pi.$$

With $n_1 = \cos\theta$ and $n_2 = \sin\theta\cos\phi$, **S2 second moments.**

$$\langle n_1^2\rangle_{S^2} = \frac{1}{4\pi}\int_0^{\pi}\int_0^{2\pi} \cos^2\theta\, \sin\theta\, d\phi\, d\theta = \frac{1}{3}, \qquad \langle n_1 n_2\rangle_{S^2} = 0.$$

The cross moment vanishes because $\int_0^{2\pi}\cos\phi\, d\phi = 0$.

For $S^3 \subset \mathbb{R}^4$, take

$$n(\chi,\theta,\phi) = (\cos\chi, \sin\chi\cos\theta, \sin\chi\sin\theta\cos\phi, \sin\chi\sin\theta\sin\phi), \qquad dS = \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi.$$

Then **Omega3 solid angle.**

$$\Omega_3 = \int_0^{\pi}\int_0^{\pi}\int_0^{2\pi} \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = 2\pi^2.$$

With $n_1 = \cos\chi$ and $n_2 = \sin\chi\cos\theta$, **S3 second moments.**

$$\langle n_1^2\rangle_{S^3} = \frac{1}{2\pi^2}\int_0^{\pi}\int_0^{\pi}\int_0^{2\pi} \cos^2\chi\, \sin^2\chi\, \sin\theta\, d\phi\, d\theta\, d\chi = \frac{1}{4}, \qquad \langle n_1 n_2\rangle_{S^3} = 0.$$

The cross moment vanishes because $\int_0^{\pi}\cos\theta\sin\theta\, d\theta = 0$.

Physical use. $\Omega_2 = 4\pi$ and $\Omega_3 = 2\pi^2$ set the reduced r^{-2} and bulk R^{-3} Gauss normalizations. The second moments give the δ_{ij}/d angular average used in the convective factor $-(1+1/d)$ and the pressure factor $1/d$.

B.2 Stage 002: Matter-Stress Inter-Defect Force Assembly

Part / anchor. Part II – Gravity

Claim status. Reduced / Controlled Reduction

Purpose. Assemble the inter-defect force between two drain defects from the matter (Noether) stress, earning the bilinear Q_1Q_2 structure, the reduced-3D r^{-2} and bulk-4D R^{-3} power laws, and the attractive like-drain sign, consuming the Stage 001 geometry primitives.

Status. Reduced / Controlled Reduction. Three verdict layers are carried from the earned source: the leading matter-stress sign `FORCE_ATTRACTIVE_DERIVED` (target-blind); the full sign `SIGN_RESIDUAL_QUANTUM_VCONF_MAXWELL_PROFILE` (honest residual); acceptance `PASS_WITH_NAMED_RESIDUALS`. The bilinear structure, the power laws, and the attractive sign are earned (form + sign); the overall magnitude is a calibration knob; the full sign carries named, profile/sim-deferred residuals.

Derivation. The medium obeys $P = K\rho^5$, $h = (5K/4)\rho^4$, with the barotropic identity $dP/d\rho = \rho\,(dh/d\rho)$. The matter (Noether) stress in Madelung form is

$$\Pi_{ij} = m\rho\, v_i v_j + \delta_{ij} P(\rho) + \sigma^Q_{ij}, \qquad \sigma^Q_{ij} = \frac{\hbar^2}{4m}\Big[\frac{(\partial_i\rho)(\partial_j\rho)}{\rho} - \partial_i\partial_j\rho\Big],$$

with the quantum-stress divergence contributing no net far-field force at this order. A defect draining flux Q sets up, by Gauss's law, the inflow

$$v = -\frac{Q\,\hat{n}}{\Omega_{d-1}\, r^{\,d-1}},$$

where $\Omega_2 = 4\pi$ $(d=3)$ and $\Omega_3 = 2\pi^2$ $(d=4)$ are the Stage 001 solid angles. The Bernoulli cross-pressure of the two overlapping inflows is $\delta P = -m\rho\,(v_1\cdot v_2)$. Integrating the traction $\Pi_{ij}n_j$ over a control sphere around defect 2 and using the Stage 001 second moment $\langle n_i n_j\rangle = \delta_{ij}/d$ gives the convective factor $-(1+1/d)$ and the pressure factor $+1/d$, whose sum is the total flux $-\,mNQ_2v_1$ (the $1/d$ pieces cancel). The stationary force $F_{12} = -\oint \Pi_{ij}n_j\,dS$, with defect 1's own inflow substituted, yields the two lanes.

Reduced-3D force law.

$$F_{12} = -\frac{m\,N_3\,Q_1Q_2}{4\pi\, r_{12}^2}\,\hat{r}_{12}.$$

Bulk-4D force law.

$$F_{12} = -\frac{m\,\rho_4\,Q_1Q_2}{2\pi^2\, R_{12}^3}\,\hat{R}_{12}.$$

Attractive sign. Since $F \propto -Q_1Q_2$, like drains $(Q_1Q_2 > 0)$ attract and opposite signs repel; the sign is derived from the Gauss-drain / Bernoulli / surface-integral chain, not assumed.

Physical use and named residuals. Target-blind, the inter-defect force is bilinear in the drain strengths, falls as r^{-2} (reduced 3D) and R^{-3} (bulk 4D), and is attractive between like drains. Honest scope: the overall normalization (I_F/Θ_Q/branch-profile) is a *calibration* knob, not derived; and the full sign carries the `QUANTUM_STRESS_FAR_FIELD`, `VCONF_BODY_FORCE`, and `MAXWELL_Z_GAUGE_JEXT` residuals, which are profile/sim-deferred and first-class.

Appendix C

Part III Stage Cards

C.1 Stage 003: Transverse Photon Pair and the Stray Longitudinal Mode

Part / anchor. Part III – Light

Claim status. OPEN

Purpose. Test whether the shear-surface brane carries light of the Maxwell kind. The transverse question earns a clean result – two massless photons at $c_\gamma^2 = \mu_R/\rho_{br}$; the longitudinal question earns a characterized departure – one stray second-class mode where Maxwell has a gauge-removed zero, with the tuned Maxwell locus reachable only by tuning.

Status. OPEN. FAIL-headline stage: the top-line verdict is a `FAIL_*`, but the earned content is the headline and the FAIL is the first-class characterized landing (no softening). *Earned headline:* the brane carries *light* – two massless transverse photons, with (for the first time in the program) $c_\gamma^2 = \mu_R/\rho_{br}$; verdict `PASS_TRANSVERSE_UNDISTURBED`. *Top-line departure:* `FAIL_CAUCHY_STRAY_LONGITUDINAL` – on the provenance-fixed single-medium branch the $(\nabla\cdot u) + \theta$ sector carries one stray propagating longitudinal DOF, a Dirac–Bergmann *second-class* pair rather than a Maxwell first-class Gauss gauge-removal. *Reachability:* the tuned Maxwell locus ($K_\theta = C_J^2/\rho_{br}$, $B_{\text{eff}} = 0$, $m_\theta^2 = 0$) is reachable and emits `C5_RESOLVED_MAXWELL_BY_TUNING` – so the FAIL is not hardcoded – but `BY_TUNING`, not `WITH_PROVENANCE`.

Derivation. The primitive brane Lagrangian (no pre-completed square used as input) is

$$L = \tfrac{1}{2}\rho_{br}(\partial_t u)^2 - \tfrac{1}{2}\mu_R(\nabla\times u)^2 - \tfrac{1}{2}B(\nabla\cdot u)^2 + J(\partial_t\theta)\,\delta\rho_B + \tfrac{1}{2}K_\theta(\nabla\theta)^2.$$

Transverse (earned). For a polarization $\perp$ $\mathbf{k}$, $\nabla\cdot u = 0$ and the θ couplings vanish, so $L_T = \frac{1}{2}\rho_{br}(\partial_t u_T)^2 - \frac{1}{2}\mu_R k^2 u_T^2$, giving

$$\omega^2 = \frac{\mu_R}{\rho_{br}}\,k^2, \qquad c_\gamma^2 = \frac{\mu_R}{\rho_{br}},$$

two polarizations $\Rightarrow$ physical_dof = 2, massless. The speed is able-to-fail: $\epsilon = 2\rho_{br}$ shifts it to $\mu_R/(2\rho_{br})$ (`FAIL_TRANSVERSE_DISTURBED`). **Longitudinal (departure).** Number conservation slaves $\delta\rho_B = -\rho_{B0}\nabla\cdot u$; integration by parts gives the Maxwell-form cross-term with the *derived* sign $C_J = -J\rho_{B0}$, and the finite conjugate-density stiffness the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$. Since $\partial_t\theta$ enters only through the first-order cross-term, the momenta are $p_u = \rho_{br}\partial_t u_L$, $\pi_\theta = Jk\rho_{B0}u_L$, forcing the primary constraint $\Phi_1 = \pi_\theta - Jk\rho_{B0}u_L$; preservation gives $\Phi_2 = -k(Jp_u\rho_{B0} + k\kappa_{\text{phase}}\rho_{br}\theta)/\rho_{br}$, and

$$\{\Phi_1, \Phi_2\} = \frac{k^2(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})}{\rho_{br}} \neq 0$$

$\Rightarrow$ second-class pair (0 first-class, 2 second-class), $N_{\text{phys}} = (4-2)/2 = 1$. The finite-stiffness pole $\omega^2 = k^2\kappa_{\text{phase}}\rho_{B0}^2/[\chi_c(J^2\rho_{B0}^2 + \kappa_{\text{phase}}\rho_{br})]$ has positive residue and a bounded reduced Hamiltonian: a real stray longitudinal mode, not a ghost and not a gauge-removal.

Physical use, fidelity, and the departure. The brane carries light: two transverse photons at $c_\gamma^2 = \mu_R/\rho_{br}$ from the medium's own moduli (light = in-plane brane shear). Honest scope: the order-parameter phase θ does *not*, on its frozen definitions, supply a gauge-removed Maxwell scalar – it leaves one stray second-class longitudinal mode; the obstruction is pinned to the Cauchy modulus $B_{\text{eff}} = \rho_{B0}^2/\chi_c$ plus the conventional phase-gradient sign $K_\theta = -\kappa_{\text{phase}} < 0$ (vs the electric sign $K_\theta = C_J^2/\rho_{br} > 0$). Fidelity: the decisive sign $C_J = -J\rho_{B0}$ is *derived* here by an in-script integration by parts (the earned source had asserted it), closing that stage's known `FIDELITY_ISSUES`; the conventional K_θ sign is a labeled postulate whose verdict-dependence is exercised by the reachable Maxwell-locus control. Resolution of the stray mode is calibrated / sim-deferred; μ_R, ρ_{br} are calibrated inputs.

Appendix D

Reproducibility Map

D.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's reproducibility rules.

D.2 Reproducibility Layers

Layer	What it controls	Minimum evidence required
No reproducibility layers populated yet.		

D.3 Canonical Source Registry

Placeholder for source artifacts.

D.4 Primary Symbolic-Audit Registry

Placeholder for symbolic audit artifacts.

D.5 Repository Freeze Protocol

Placeholder for future release-freeze requirements.

Appendix E

Source File Index

E.1 Purpose of This Appendix

Placeholder for the rebuilt ledger's source-file index.

E.2 Source Classes

Placeholder for source classes.

E.3 Generated Ledger Files

File family	Role
`paper/frontmatter/*.tex`	Placeholder governance chapters.
`paper/parts/*.tex`	Future theorem-block chapters.
`paper/stages/stage_NNN.tex`	Future per-stage cards.
`scripts/ledger_stageNNN_*.py`	Future SymPy audit scripts.
`mathematica/ledger_stageNNN_*.wl`	Future Mathematica audit scripts.

E.4 Precedence and Conflict-Resolution Rules

Placeholder for source precedence and conflict-resolution rules.

Appendix F

Layer-2 Fit-Insertion Index

F.1 Generated Coverage Summary

Item	Count
Phase-A candidates	0
Audited candidates	0

F.2 Phase-A Candidate Rows

No candidate rows exist in the empty ledger skeleton.

F.3 Visible Grouping and Exclusion Accounting

No grouping or exclusion accounting exists yet.

Appendix G

Layer-2 Parameter-Provenance Audit

G.1 Generated Coverage Summary

Item	Count
Top-level provenance parameter records	0
Canonical candidates	0

G.2 Published-Target Benchmark Sources

No benchmark rows exist in the empty ledger skeleton.

G.3 Parameter Genealogy Rows

No parameter genealogy rows exist yet.

G.4 Visible Alias Accounting

No alias accounting exists yet.

Appendix H

Fill Workflow and Editorial Rules

H.1 Purpose of This Appendix

Placeholder for the rebuilt ledger fill workflow.

H.2 What Filled Means

Placeholder for completion criteria.

H.3 Governing Maps Used During Filling

Placeholder for status, notation, anchor, dependency, source, and reproducibility controls.

H.4 Stage Appendix Fill Protocol

Placeholder for future stage-card authoring rules.

H.5 Theorem-Block Fill Protocol

Placeholder for future Part authoring rules.

H.6 Minimal Build and Static-Check Workflow

1. Confirm every retained `\input` target exists.
2. Run the SymPy and Mathematica audit runners.
3. Build `paper/pde_ledger.tex`.